An ensemble approach to enhance accuracy and robustness of persistent malware detection in-vehicle infotainment system

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***Abstract*— The growing integration of information technology into vehicles makes the importance of automotive cybersecurity protection systems necessary. There are many kinds of attacks that threaten the safety of vehicles and the privacy of their owners. One such possible malicious action is remote code execution, which allows an attacker, for example, to exploit the software of an automobile to control an automobile's function s such as brakes and steering. These consequences include risk to safety, infringement of privacy, economic losses, damage to the automobile, and spoiling reputation for the automobile industry. The continual malware-detecting model based on voting ensemble classifiers has better accuracy than other machine learning models specifically on infotainment system. It deals with real-time threats on the Internet of Things devices. The proposed model provides a comprehensive view of how an attacker might target a system, mapping out potential entry points, data flows, and trust boundaries in automated vehicles.**

Keywords*— *Vehicle infotainment system, Threat Analysis, Malware Threats, Voting Ensemble Classifier.

I. INTRODUCTION

The automotive industry is undergoing a rapid transformation, driven by advancements in technology, particularly in the realms of connectivity and autonomy. While these innovations offer immense potential, they also introduce significant cybersecurity challenges. This literature review delves into the existing body of research on automotive cybersecurity, examining key threats, vulnerabilities, and mitigation strategies [1]. This includes malware, hacking, unauthorized access, data breaches, and physical attacks on vehicles. Exploring how attackers exploit vulnerabilities in vehicle systems, networks, and software. It is very important to understand emerging threats such as ransom ware, denial-of-service attacks, and supply chain vulnerabilities. In this work main focus is done on persistent malware threats and its impact on vehicle infotainment application. In the Fig 1. below shows possible ECU components which are vulnerable to cyber-attack. The other attack points in vehicle are perimeter device such as In Vehicle infotainment, telematics control unit, diagnostic port, and back-end server which can be cloud based service, internal vehicle communication, Key FOB, Smart Phone Apps and external attacks. In this paper focus mainly on smart phone and show case different types of cyber-attacks and its type of threats.

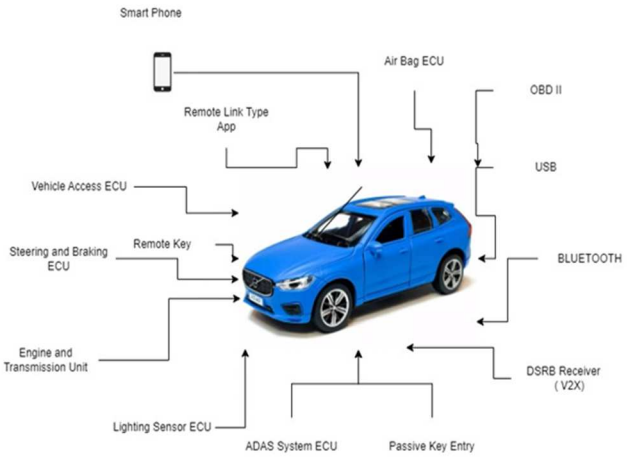
The focus is risk assessment to identify potential ransomware threats and assess their impact. It's very important to implement robust security measures to prevent ransomware attacks on infotainment system. The detection employs advanced detection technologies to identify ransomware early [2]. The development of comprehensive incident response plans. and effective data backup and recovery strategies. The foster collaboration between government, industry, and academia. Its clear ethical guidelines for ransomware response, including ransom payments most important payload in terms of persistent threat.

To mitigate these risks, vehicle owners should adopt several strategies such as regular software updates ensures that security patches are applied promptly but many users are unaware, while strong password policies protect accounts linked to the vehicle and utilizing secure Wi-Fi connections and limiting Bluetooth connectivity to trusted devices can also minimize exposure to threats, knowledge using USB drives from untrusted sources, which can lead to persistent malware threats which are major factors to be considered [3]. There is a limited resource for security mechanism and performance constraints to meet real world requirements. The types of attacks which are possible are spoofing attack, data replay, Denial of Service, Tampering, Side Channel Attack, Man-in-Middle and Jamming attacks. Spoofing in Cybersecurity is when an attacker pretends to be authorized user in an attempt to gain our confidence, get access to our vehicle systems, steal Data, or spread malware [4]. Tampering refers to deliberate and unauthorized manipulations of vehicle components. Tampering involves certain intervention and changes to be performed on the vehicle. The tampering techniques may include program/data flashing and connecting emulation devices leads to payload.

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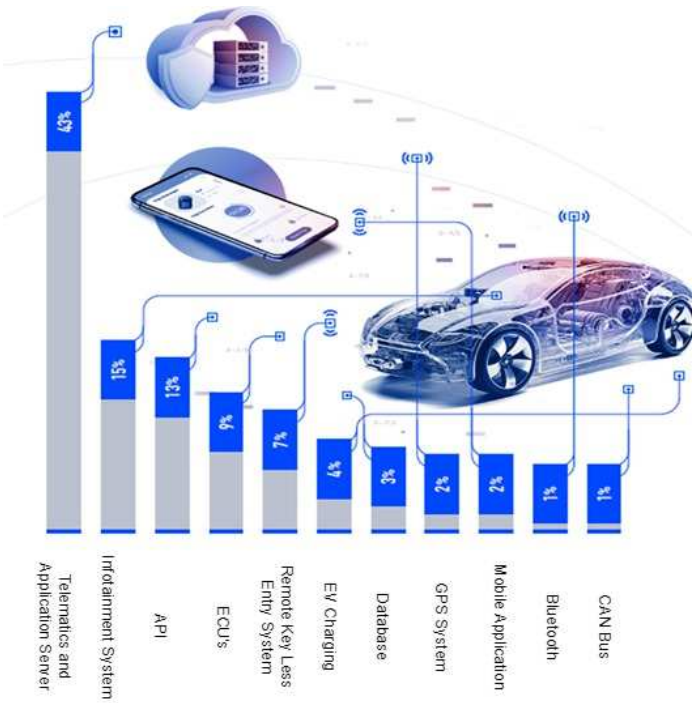
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Fig 1. ECU components vulnerable to cyber attacks



This work has three different parts Risk Analysis, Assessment and building model using machine learning algorithm based on data sets collected from various repositories. The Fig 2. below shows automotive cybersecurity landscape, incidents and attack actor. The challenges with current technologies are vehicle complex with various types of attack surfaces, connectivity, new features and architecture increased in modern vehicles. The existing technologies are not designed with cyber-attack mind. It is meaningful transfer of existing IT security to new development. Although there are plenty of attacks our aim is to mainly focus on ransomware attacks which are possible through smart phone [5].

Fig 2. Automotive cybersecurity landscape, incident and attack vector



Ransomware, traditionally targeting computers, is increasingly targeting IoT devices. This trend is driven by vulnerable devices as many IoT devices have weak or no security measures [6]. The data value with IoT devices often collect valuable data, making them attractive targets. The extortion potential leads to disrupting critical IoT services and significant financial losses.

II. RELATED WORK

A. *Cybersecurity Ecosystem*

The rapid advancement of automotive technology, particularly in the realm of automated vehicles (AVs), has significantly increased the complexity of the vehicle system. This complexity, coupled with the increasing connectivity of vehicles, has introduced unprecedented cybersecurity challenges. The Automotive Safety Integrity Level (ASIL) focus primarily on physical hazards, neglecting emerging cyber threats. This duality creates a gap that needs to be addressed by expanding such frameworks as it pertains to cybersecurity. Traditionally, safety risks revolved around hardware and system failures which could be easily anticipated [7]. It The scale of destruction that a cyberattack has the potential to create makes measuring its impact nearly impossible, further complicating risk mitigation strategies. A system can be designed to be functionally safe yet remain easily penetrable by cyber threats. New cybersecurity integrity levels for the automobile industry is safety and security of automated vehicles. There are inadequacies existing frameworks to strengthens the automotive industry's ability to counter cyberattacks [8].

B. *Attack-surface in automated vehicle*

A Cyber Physical System Attack Surface represents the quantitative metric evaluating the vulnerabilities of automotive systems to possible cyber-attacks. Thus, it is possible to assign value to a system depending on its vulnerability level. This index aids in effectively allocating the required level of attention needed to be given to security, classifying critical features, and analyzing the complete security level of a vehicle. System modelling Attack ability index is greatly influenced by the system modelling, threat identification, vulnerability assessment, attack path analysis, index calculation and risk prioritizing processes. [9] False data injection attacks are an ever-growing problem for connected and automated vehicles (CAVs). These kinds of attacks pertain to the control and safety of a devised system as these involve altering the information sent to the vehicles from the infrastructure or vice versa. By the use of a cloud, real time data concerning CAVs like sensors, the vehicle's dynamics, as well as V2X data is gathered and sent to the cloud. For CAVs, sandboxing provides a new way of detecting and preventing such attacks in real time [10]. Anomaly detection refers to the identification of irregular patterns or variations and their corresponding change from expected behavior. Verification on the basis of the model entails is checking information against the set vehicle models and simulation. [11]

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C. Real-time monitoring in automated vehicle

This framework involves collecting, processing, and analyzing vast amounts of vehicle data to identify potential issues, predict maintenance needs, and optimize vehicle operations. Various sensors are installed in the vehicle to collect data on engine performance, tire pressure, brake system, emissions, and other critical parameters. Initial data processing and filtering occur within the vehicle to reduce data volume before transmission. Processed data is securely transmitted to a cloud-based platform for storage and analysis [12]. Data is encrypted both during transmission and storage to protect sensitive information. Robust access controls are implemented to restrict unauthorized access to data. Continuously monitor data for anomalies that could indicate security breaches. Apply machine learning algorithms to identify patterns, trends, and anomalies in the data. Develop models to predict vehicle failures, maintenance needs, and optimal performance conditions [13]. The Real-time Monitoring has continuously monitor vehicle health parameters for abnormal conditions. The Diagnostic Alert having generate alerts for potential issues, enabling timely maintenance. The Predictive Maintenance is able to predict maintenance needs based on data analysis, optimizing costs and vehicle uptime. The driving behavior analysis where driving patterns are to identify areas for improvement and provide personalized recommendations. The fuel efficiency optimization maintains optimize fuel consumption through data-driven insights. The Route Optimization suggest optimal routes based on traffic conditions and vehicle performance. [14]

D. Threat Analysis and Risk Assessment (TARA)

In-vehicle networks, primarily dominated by CAN bus, are increasingly becoming targets for cyberattacks due to their lack of inherent security mechanisms. Traditional intrusion detection systems often struggle to accurately pinpoint the attacker, leading to ineffective countermeasures. The anomaly detection involves monitor network traffic for deviations from normal patterns in terms of message frequency, length, and content. Employ algorithms like Support Vector Machines (SVM), Random Forests, or Neural Networks to identify unusual traffic patterns [15]. The attack classifications are behavioral analysis and pattern recognition. The attacker fingerprint involves message characteristics, network behavior and machine learning. The real time response involves with rapid detection, countermeasures and isolation. The important challenge is balancing detection accuracy with minimum false alarm. [16] The combination of the "Asset Container" method and CWSS (Common Vulnerability Scoring System) offers a structured approach for conducting threat analysis and risk assessment (TARA). The asset container method focuses on identifying assets and their potential vulnerabilities, categorizes assets into containers based on their functions or locations, helps in understanding the attack surface and potential impact. CWSS quantifies vulnerability severity based on factors like attack complexity, required privileges, user interaction, and impact [17]. This provides a standardized metric for comparing vulnerabilities. By integrating these methods, in this work can systematically identify, assess, and prioritize risks. By addressing these challenges and leveraging the strengths of the asset container method and CWSS, can effectively manage cybersecurity risks and protect their assets. [18]

Extracted features are transformed into a suitable format for machine learning algorithms, often involving normalization, scaling, and dimensionality reduction techniques. An appropriate algorithm (e.g., Random Forest, Support Vector Machine, and Neural Network) is trained on a dataset of labelled memory samples (benign and malicious). The challenges faced are trained model analyses incoming memory data to classify it as benign or malicious, real-time memory analysis can be computationally intensive, requiring efficient algorithms and hardware. Ransomware is constantly evolving, necessitating continuous model updates. [19]

The challenges faced are trained model analyses incoming memory data to classify it as benign or malicious, real-time memory analysis can be computationally intensive, requiring efficient algorithms and hardware. The persistent payload such as backdoors, adware's and spywares are constantly evolving, necessitating continuous model updates [20]. The literature analysis concludes by highlighting the variety of methods used to identify backdoors and other persistent threats, with machine learning algorithms like random forest being a popular option because of their ability to achieve high accuracy rates. In order to combat the constantly changing threat landscape of these payloads, these strategies emphasize a number of different areas, such as feature selection, behavioral profiling, and static analysis [21].

III. PROPOSED METHODOLOGY

Threat Analysis and Risk Assessment (TARA) is a process that integrates all phases of evaluation identification, analysis, and evaluation of possible threats and vulnerabilities relevant to an organization's security. Estimation of probabilities and the total possible outcomes value of occurrence for these threats is likely to be assessed. Method of risk evaluation concerning asset and asset-critical infrastructure identification proposing relative high priority asset and threat mitigation frameworks. This framework uses the concept of big data analytics to maximize performance, safety, and cost-efficiency of the vehicle [22]. It incorporates the collection, processing, and analysis of large volumes of data to identify problem areas, maintenance forecasting, and optimizing operations. The installation of various sensors into the vehicle permits the acquisition of data regarding the engine, tire pressure, the brake system's performance, emission processes, and other important constituents. Initial filtering and processing of data occurs within the vehicle based on given thresholds to constrain the amount of data transmitted specifically on infotainment system [23]. The filtered data is sent to data clouds in a protected manner, stored, and analyzed. The sensitive data that relies on advanced encryption during transmission and storage is additionally, protected from unauthorized access. Effective restrictions to prevent unauthorized users from receiving the stored data are used. A framework related to the method of quantitative risk assessment and evaluation which determines the probability and consequences of risk using statistical approaches [24].

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A. Online attacks in Vehicles

With the growth of electronic components in the car, cyberattacks have become one of the paramount concerns. As vehicles are fitted with more technology such as wireless network connections and internet enabled infotainment systems, it increases the risk of an attack. growing connectivity of the modern automobile has made new attacks possible, so it's never been more important for car owners to be aware of the variety of cyber threats they're exposed to. Cyber criminals could exploit the breeches to access a car's systems remotely and gain control of key functions, including brakes, steering and even the engine. This type of invasion creates severe threats, as a hacked vehicle may cause accidents and put not only the driver and passengers but also pedestrians at risk. Apart from bodily injury, theft of data is also a severe issue; confidential data such as personal information, driving behavior, and current location are stolen, causing identity theft and bank fraud. Cybercriminals can also initiate Denial of Service (DoS) attacks [25], flooding a vehicle's systems with malicious traffic and interfering with critical operations like navigation and emergency services. Ransomware and other persistent payload attacks pose an added threat, as hackers can lock down a vehicle's information and ask for a ransom to restore it.

With technology getting better and better, it is important for vehicle manufacturers and consumers alike to be aware of cyber threats. Knowing the threats and taking measures proactively safeguards ourselves and our vehicles against online attacks specifically on infotainment system.

B. Data Collection

The data collected from open-source repositories which includes collection of backdoor attacks done on vehicles specifically on infotainment applications. In order to build machine learning model for classification of attack few benign applications in equal proportion also collected from CIC MalDroid 2020, CCCS-CIC-AndMal2020, Kaggle and IDN malware data set collected over period of six months and tested using hybrid approach shown in table below [22]. The real time testing of data done with support of test beds given by K7 computing and Volvo Groups. The datasets are processed with broader spectrum of inputs, making it more robust to variations and noise in real-world data applied on Android play infotainment system in automated vehicles shown in Table 1.

TABLE I. DATA COLLECTED FROM REPOSITORIES

S. No	Sample	Count
1	Backdoor	1050
2	File Infector	669
3	Ransomware	1072
4	SMS Attack	1025

C. Classification of Malware using ML Approach

Machine Learning (ML) has become a powerful tool for detecting and classifying Android malware. By analyzing the behavior and characteristics of malicious apps, ML models can accurately identify threats and protect users from harm. Permissions requested by applications can serve as an early warning sign of potential malicious behavior. For instance, apps that seek excessive permissions, such as accessing contacts or sending SMS messages, may raise red flags. Additionally, the API calls made by an app provide insight into its functionality; malicious apps typically exploit APIs related to network communication, data storage, or system-level interactions to execute harmful actions. The opcodes within an app's byte code also play a significant role in identifying its behavior; certain opcodes can indicate intentions aligned with malicious activities. The proposed voting ensemble classifier is shown in Fig 3. below.

For evaluating the safety of an app, static and dynamic analysis techniques are utilized. Static code analysis entails studying the source code of the app for bugs and unusual patterns, whereas dynamic analysis monitors the app's runtime behavior to analyze its interaction with the operating system and other programs. Machine learning (ML) algorithms can also be used to augment the detection of malicious apps through supervised and unsupervised learning methods. Under supervised learning, the classification algorithms like Random Forest, Support Vector Machines (SVMs), Naive Bayes, and Decision Trees are used, with feature extraction algorithms like Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF).

Unsupervised learning techniques, such as clustering algorithms like K-means and hierarchical clustering, and dimensionality reduction like Principal Component Analysis (PCA) and t-SNE, can discover hidden structures in data without labels. Yet, there are some issues and considerations in using these ML methods. The dynamic behavior of malware requires ML models to be frequently updated to keep their relevance against new threats. Furthermore, feature engineering is an essential factor that demands precise choice of meaningful features to provide correct classification, especially while handling sophisticated malware. Ensuring that training ML models with high-quality datasets is also crucial; however, collecting and labelling large datasets can take a lot of time. Finally, trade-offs in performance need to be taken care of since different ML methods have different characteristics in terms of accuracy, speed, and interpretability, and thus selecting the proper technique based on certain requirements is important in automated vehicle infotainment system.

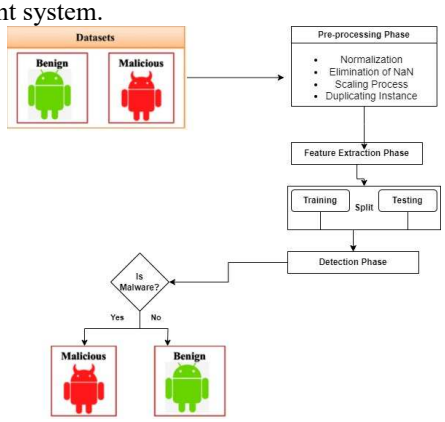


Fig 3. Proposed Voting Ensemble Classifier

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D. Ensemble approach in Malware Classification

Ensemble methods are an effective machine learning technique that involve combining many models together to enhance predictive accuracy. Such methods can be especially valuable in malware classification, where the variety and sophistication of malware may hinder individual models from being highly accurate. In this method boosting technique is used where weighted aggregation of prediction of all the models is performed. While cybersecurity will continue to be a top priority, having strong security measures in place will be necessary to protect against cyberattacks and unauthorized access to data. Secure communication channels will be the main focus to ensure data integrity for the communication between vehicles and external networks. In addition, the combination of infotainment systems with autonomous driving capability will provide a smooth experience to passengers.

$$y = \text{argmax}(\Sigma(Pi(x, c))) \quad (1)$$

Pi(x, c) is the probability assigned by the i-th base model to class c for the data point x. argmax returns the class label c that maximizes the sum of probabilities shown in equation 1. The advantages of ensemble method acquire greater significance in machine learning because they can greatly increase accuracy, especially on difficult tasks like malware detection. By combining multiple models' predictions, the methods can take advantage of the strengths of one model while covering for the other's weaknesses. The collective method not only results in greater overall accuracy but also plays an important part in preventing overfitting. Overfitting results when a model is over-specialized for the training data and hence performs badly on new, unseen data. Ensemble methods reduce this probability by combining different models that are diverse, and thus ensure that predictions don't become too specialized. Ensemble methods also show greater resistance to noise and outliers in the data. The predictions from different models combined can successfully suppress the individual errors, leading to more stable predictions. The heterogeneity of base models is partly responsible for the robustness, because each model is likely to perceive the data differently and can provide different insights to enhance the process of classification. Ensemble methods therefore not only enhance predictive accuracy but also create a more stable and flexible methodology to handle complex classification tasks handling persistent payloads which can target Android play infotainment system.

IV. RESULTS AND DISCUSSION

The principal components are often correlated, making it challenging to understand the individual contributions of each component to the variance explained from my previous findings [21]. Ensembles can act as a form of regularization, preventing individual models from overfitting the training data. Ensembles can be more robust to noise and outliers in the data, as individual models may be affected differently by these factors. In this approach multiple models by bootstrapping different subsets of the training data and averaging their predictions. By combining multiple models, ensembles can leverage the strengths of each individual model and mitigate their weaknesses, leading to better overall results. The table below shows comparison of individual model with ensemble approach achieving better accuracy for prediction of malware payload under vehicle infotainment applications built with Android play features. The tables below show results obtained from individual classifiers based on different types of payloads. The table 2 below shows results from random forest model considering important factors from performance evaluation such as precision, recall and F1 score comparison done with multiple class label such as backdoor, file infector, ransomware and SMS attack. VEM_Acc is a fundamental metric that measures the proportion of correctly classified malware samples out of the total number of samples [34]. It is calculated as shown in equation 2.

$$VEM_Acc = \frac{X+Y}{X+Y+Z+A} \quad (2)$$

The ‘X’ is correctly classified malware instances. The factor ‘Y’ defines correctly classified benign instances. The ‘Z’ defines benign instances classified as malware and ‘A’ defined malware instances misclassified as benign. VEM_Prec, VEM_Prec, and VEM_score are key performance metrics that evaluate the classifier's ability to correctly identify malware threats shown in equations below. VEM_Prec measures the proportion of correctly identified malware samples among the total samples classified as malware. VEM_Prec measures the proportion of correctly identified malware samples among the total samples classified as malware. VEM_score is the harmonic mean of precision and recall, providing a balanced measure of both.

$$\text{VEM_}Prec = \frac{X}{X+Z} \quad (3)$$

$$VEM_Rec = \frac{X}{X+A} \quad (4)$$

$$VEM_score = 2 * \frac{VEM_Prec * VEM_Rec}{VEM_Prec * VEM_Rec} \quad (5)$$

The VEM_ROC curve illustrates the trade-off between the VEM_TPR and VEM_FPR at various threshold levels.

$$VEM_TPR = \frac{X}{X+A} \quad (6)$$

$$\text{VEM_}FPR = \frac{A}{A+Z} \quad (7)$$

The VEM_AUC is a summary metric that quantifies the classifier's ability to distinguish between malware and benign samples. A higher VEM_ AUC value (closer to 1) indicates superior classification performance. Table 2 below presents the performance comparison of individual classifiers (Random Forest, SVM, k-NN, and Gradient Boosting) and the proposed voting ensemble classifier model. The Fig 4. shows comparison parameter of truth and predicted of proposed classifier using heatmap.

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TABLE II PERFORMANCE COMPARISON

VEM_CLASSIFIER	VEM_ACC (%)	VEM_PREC (%)	VEM_REC (%)	VEM_SCORE (%)	AUC
RF (Ahmed et.al ,2023)	95	93	96	94	0.95
SVM (Sana et.al, 2024)	94	92	93	93	0.94
k-NN (Lakhan et.al, 2024)	94	94	94	94	0.94
Gradient Boosting(Coosa et.al, 2024)	94	94	94	94	0.94
Voting Ensemble (Proposed Work)	**96**	**96**	**96**	**96**	**0.96**

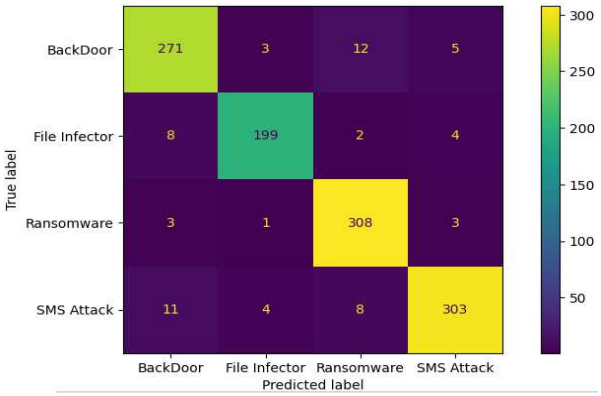


Fig 4. Heatmap comparison for proposed ensemble classifier

Ensembles often involve more parameters and a more complex architecture than single models, which can lead to longer training and deployment times, especially for large datasets. During prediction, all individual models in the ensemble make their predictions, and then these predictions are combined averaged and voted. This aggregation step adds to the inference time, making ensemble models generally slower for real-time predictions compared to a single, simpler model. Fig 5. below illustrates the VEM_ROC curves for the individual classifiers and the ensemble model, highlighting the comparative performance in distinguishing malware threats.

The threats in vehicle infotainment applications are growing concern as most of them uses Android play device, and it's no surprise given how popular the platform is and the open nature of its ecosystem. This widespread issue puts users at risk of serious problems like data theft, financial losses, identity fraud, and even damage to their devices [30]. It's unsettling to think that malware can lurk in places we often trust, such as apps from unofficial sources, compromised websites, and occasionally even the Google Play Store itself, which we typically consider safe. There are various types of malwares to be aware of, including viruses,

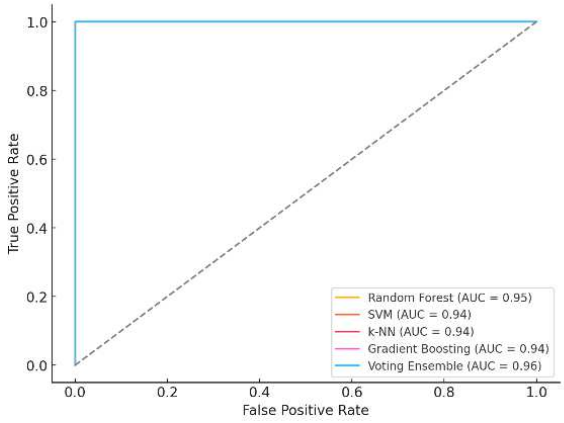


Fig 5. Threshold level for proposed classification models

worms, Trojans, ransomware, spyware, and adware. Which are focused mainly in result section. Each of these persistent threats are labelled with multiple class and given as input model. The proposed ensemble classifier has better accuracy compared with other machine classifiers and suitable for sustainable urban growth focusing on security aspects as shown in Table 2.

V. CONCLUSION AND FUTURE WORK

The proposed voting ensemble classifier model will help system to achieve higher levels of abstraction. More advanced voice assistants with improved natural language processing capabilities will enable more complex interactions, and biometric authentication techniques such as facial recognition will ensure safe access and personalize experiences for unique users. As cars enter autonomous mode, infotainment systems will offer entertainment choices but also enhance overall passenger experience through personalized content and services tuned to individual user preferences. It is important to make better user experience with key priority which include gesture control, voice command, and even eye-tracking for effortless interaction. Through these primary areas, the development of automobile infotainment systems will finally reshape the driving experience, creating vehicles as connected, tailored, and safe mobility solutions of the future. The model's resilience to adversarial attacks or obfuscation can be an area for future exploration on infotainment system in automated vehicle.

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